

Google Fiber

Repeat Call Analysis Dashboard

How I built a full BI pipeline — from stakeholder requirements to a live Tableau dashboard

BigQuery & SQL

ETL Pipeline

Tableau

Dashboard

Google BI Certificate

Capstone

The Problem

Google Fiber's customer service team was flying somewhat blind. They knew customers were calling back after their first contact, but they had no structured way to measure how often this was happening, which issues were driving it, or whether one market city was performing worse than the others.

Repeat calls are expensive. Every callback that could have been resolved on the first interaction is a failure of the support system — it costs the business time, it frustrates the customer, and it inflates call volume for no added value. Leadership wanted a dashboard they could use to monitor this pattern and drive operational improvements.

The core questions:

- How often are customers calling back after their first contact?
- Which problem types generate the most repeat calls?
- Which market city has the highest repeat call volume?

My Role

This was a solo end-to-end BI project completed as the capstone for the Google Business Intelligence Professional Certificate. I was responsible for every stage: requirements gathering, pipeline design, SQL development, and dashboard delivery.

The project was framed as a mock interview process — I was acting as a BI Analyst candidate presenting work to a fictional Google Fiber customer service team. That framing actually made the stakeholder requirements documents feel more real, because I had to think about what the stakeholders actually needed rather than just what the data showed.

My Approach

Stage 1 — Requirements & Planning

Before touching any data, I completed three BI planning documents based on stakeholder interview notes: a Stakeholder Requirements Document, a Project Requirements Document, and a Strategy Document. This stage forced me to get precise about what the dashboard needed to answer and who it was for.

One requirement that stood out: Lead BI Analyst Minna Rah specified that the dashboard needed to be accessible — large print and text-to-speech alternatives. That became a non-negotiable constraint I carried through to the final build.

Stage 2 — Data Pipeline (BigQuery)

The data arrived as three separate CSV files, each representing a different anonymised market city (market_1, market_2, market_3). I uploaded all three to Google BigQuery and wrote a UNION ALL query to combine them into a single reporting table:

```
SELECT * FROM `project.dataset.market_1`  
UNION ALL  
SELECT * FROM `project.dataset.market_2`  
UNION ALL  
SELECT * FROM `project.dataset.market_3`
```

The tables shared identical schemas, so no transformation was needed beyond the UNION ALL. The combined output (1,350 rows) was exported as combined_calls.csv and imported into Tableau.

This stage is where my interest in data engineering really kicked in. The pipeline work — understanding how data moves from sources into a reporting layer — was more interesting to me than the visualisation that came after it. It's what pushed me to go deeper into DE concepts beyond what the certificate covered.

Stage 3 — Dashboard (Tableau)

I built the dashboard in Tableau using the combined_calls target table. The dataset structure was slightly unusual — repeat calls were recorded as separate columns (contacts_n_1 through contacts_n_7), each representing calls made that many days after the first contact. I had to account for this when calculating total repeat calls and building the behaviour curve.

Charts built:

- KPI summary cards for total first contacts and total repeat calls
- Trend lines for repeat calls and first contacts over time (Jan–Mar 2022)
- Repeat Behaviour Curve — showing the rate of callbacks across the 7-day post-contact window
- Horizontal bar chart of repeat calls by problem type (with human-readable aliases, not raw type codes)
- Bar chart of repeat calls by market city
- First contact volume by day of week

The problem type aliases were one of those small things that matter a lot in stakeholder-facing work. The raw data used type_1 through type_5. Presenting those codes in the dashboard would have been useless to a non-technical audience. I mapped them to their actual categories using Tableau aliases before publishing.

Results

64,939

Total First Contacts

20,240

Total Repeat Calls

31.2%

Repeat Call Rate

- Internet & Wi-Fi issues (Type_5) drove 50.66% of all repeat calls — the single largest driver by a significant margin
- Technician Troubleshooting (Type_2) was second at 29.58% — together, these two categories account for over 80% of all repeat calls
- Market_1 generated 12,647 repeat calls vs. Market_2's 787 — a 16x gap that warrants deeper investigation into whether this is a size difference or a performance gap
- The Repeat Behaviour Curve shows the sharpest drop between Day 0 and Day 1 — most customers who are going to call back do so within 24 hours

What I Learned

The planning stage is underrated. I went in thinking the dashboard was the main deliverable. Looking back, the stakeholder requirements and strategy documents were just as important — they kept me focused on the actual business questions rather than building charts because the data was interesting.

Aliases and labels are stakeholder experience. Raw field names in a dashboard are a red flag. A chart that says “type_5” means nothing to a customer service manager. Mapping codes to readable labels is not a nice-to-have — it’s part of delivering a usable product.

The pipeline work was what I wanted more of. Writing the UNION ALL query in BigQuery, thinking about how the data needed to be structured before it could be visualised — that stage genuinely excited me more than building the charts. It’s what pushed me to move beyond BI into data engineering.

What I’d Do Differently

- Add a first-contact resolution rate metric — the current dashboard shows repeat call volume but doesn’t explicitly surface the FCR rate as a KPI, which is what leadership actually cares about most
- Build year-over-year comparison capability — the dataset only covers Jan–Mar 2022, which limits trend analysis. With more historical data I’d add a YoY overlay
- Investigate Market_1 vs Market_2 more deeply — the gap is stark but the dataset doesn’t tell us whether it’s driven by market size, problem type mix, or actual resolution quality
- Automate the pipeline — the UNION ALL was run manually in BigQuery. With Airflow or a scheduled query, this would update automatically as new data comes in